

The Development of a Dashboard Tool for Visualising Online Teamwork Discussions

Rebecca Vivian, Hamid Tarmazdi, Katrina Falkner, Nickolas Falkner and Claudia Szabo

School of Computer Science

The University of Adelaide, South Australia, Australia, 5005

Email: firstname.lastname@adelaide.edu.au

Abstract—Many software development organisations today adopt global software engineering (GSE) and agile models; requiring software engineers to collaborate and develop software in flexible, distributed, online teams. However, many employers have expressed concern that graduates lack teamwork skills and one of the most commonly occurring problems with GSE models are issues with project management. Team managers and educators often oversee a number of teams and the large corpus of data, in combination with agile models, make it difficult to efficiently assess factors such as team role distribution and emotional climate.

Current methods and tools for monitoring software engineering (SE) teamwork in both industry and education settings typically focus on member contributions, reflection, or product outcomes, which are limited in terms of real-time feedback and accurate behavioural analysis. We have created a dashboard that extracts and communicates team role distribution and team emotion information in real-time. Our proof of concept provides a real-time analysis of teamwork discussions and visualises team member emotions, the roles they have adopted and overall team sentiment during the course of a collaborative problem-solving project. We demonstrate and discuss how such a tool could be useful for SE team management and training and the development of teamwork skills in SE university courses.

I. INTRODUCTION

Many software engineering (SE) companies today adopt more open, flexible, dispersed development models that allow software engineers to work in globally or nationally distributed teams to accomplish complex software development projects [1]. As companies continue to move toward distributing software development activities offshore or across numerous company sites, global software engineering (GSE) models are adopted. GSE projects are often large scale and their ‘global’ nature leads to increased complexity, resulting in GSE as having a high failure rate, primarily caused by project management issues [2]. The flexibility and human-focused method of GSE places a greater emphasis on a need for effective planning, organisation, and management of team processes [3]. Due to the highly complex nature of software development and a reliance on teamwork, SE employers identify communication, teamwork, and problem-solving skills as critical to SE professional practice [1], [4]. However, research has identified that many new SE graduates struggle with teamwork and soft skills [5], [6], [7].

Employers urge universities to pay greater attention toward the development of students’ teamwork and soft skills in coursework and propose that this is an area that requires urgent

attention [8]. Universities can provide learning opportunities for students to engage in teamwork with group projects and problem-based learning [9], however, it has been argued that often these skills are expected to be developed through teamwork experience, rather than something that is explicitly taught and assessed [10]. Often educators are responsible for a large number of students and multiple teams, making it difficult to traverse large amounts of online conversational data to provide feedback, particularly about specific behaviours and team roles. The same issues could be said for the industry context: in what ways are SE companies assessing and training employers in teamwork and soft skills? As SE companies increasingly adopt GSE models and managers are often required to supervise multiple teams online, it becomes increasingly difficult for employers to monitor online teamwork, particularly across distributed environments and multiple platforms.

In this paper, we review the methods and tools for monitoring and assessing online teamwork at university and in industry. We follow by presenting the development of our teamwork dashboard, which has been built upon our early work of team role analysis [20] and recent developments in learning analytics. We use student discussions from a final-year SE collaborative assignment as an example to demonstrate the features of how the dashboard can be utilised. Lastly, we conclude with future research directions and possible applications of the dashboard for university learning and industry team performance management.

A. Software Engineering Teamwork

Software companies are increasingly adopting online modes of teamwork, such as GSE, in order to allow employees to collaborate on software development projects with colleagues at other sites or in different locations [2]. In fact, a portion of software industry today have shifted all, or part, of their software development online. These online SE teams may be geographically, organisationally, and/or time dispersed workers brought together with online tools [12] that allow them to collaboratively complete complex SE projects. GSE is often large scale, leading to increased complexity, particularly with the aspect of it being ‘global’ [2].

SE team success may be influenced by a number of factors, such as individual contributions, the context, and the history of team members [13]. Successful teams are identified as being agile, self-organised, have clearly defined objectives,

equal member participation [10] and are able to effectively monitor and evaluate team progress [14]. In GSE, many of these aspects that can contribute toward team success can also see the demise of software projects, with team conflicts, cohesion, knowledge-sharing and communication processes all placing significant strain on the team members and managers, if not working well [3]. With GSE approaches, there is an increased complexity in terms of human-related activities and the monitoring of team performance. It has been reported that one of the primary causes of failure in GSE is due to project management issues [2]. Therefore, SE companies are paying greater attention to project management tools and employee skills that can support effective teamwork and successful product development [3].

Agile software development processes are increasingly popular in organisations [15], but the self-organising nature of the method and lack of strictly allocated roles means that graduates need to know how to transition between roles in order to adopt new roles as their team requires. Leadership is not usually dispersed evenly among self-organised team members and is not necessarily fixed, as leadership may alternate between individuals, depending on who has the specific abilities and skills required for the team at a point in time [16]. In self-organising teams, it has been found that members often display spontaneous role behaviours and, despite the omission of a designated leader, planning and leadership behaviours still emerge [17]. However, it has also been recognised that a lack of coordination behaviours exist in agile teams, in comparison to teams with allocated roles [16].

There is a great deal of pressure on universities to produce graduates, to not only have the technical skills and theoretical knowledge but to also have the skills required to participate as an effective team member.

B. University Teamwork

In Computer Science (CS) programs, students may be scaffolded through various team roles using the pedagogical approaches, similar to Contributing Student Pedagogy (CSP) [9], [18]. An educator may model particular roles or allocate roles to students in collaborative tasks to encourage role adoption and transition between roles [9]. In studies of SE teamwork at university, it has been recognised that students struggle with producing software designs in teams [19] and when asked to evaluate peers' teamwork skills, 'collaborative skills' are among the lowest scoring teamwork attributes [10]. Further, many students express a dislike of teamwork because of negative experiences with the inequality of member participation [20], the presence of perceived 'slackers' [21] and pressure to complete a majority of the workload when peers are not contributing [20].

Teamwork assessment at university does not necessarily favour students in these situations as assessment usually focuses on the product and technical knowledge presented, rather than on the processes and assessment of skill development [13]. Teamwork skill development in CS is something that educators tend to hope students will develop through 'experience'

[10] by partaking in a CS degree. As a result, little attention is focused on teamwork, participation and management that individual members practice during collaborative tasks and how teamwork is assessed and improved.

Software is essentially a product of the cognitive processes of individuals engaged in innovative teamwork [3]. As both SE student and industry teams are increasingly required to collaborate with others in distributed locations, much of the collaboration is now occurring online with various tools, meaning that they are leaving traces of their teamwork processes online. Such data provides an opportunity for the development of innovative tools that analyse and visualise such discussions which could be used both in academia and industry contexts.

II. TOOLS FOR TEAMWORK ANALYSIS

It is recommended [22] that software tools for GSE should help to alleviate problems such as: (a) *Geographic Dispersion*, which sometimes causes a loss of synchronous communication or team interactions, since the sites are in different time zones; (b) *Control and Coordination Breakdown*, owing to the difficulties created by a distributed environment; (c) *Loss of Communication*, with communicating online instead of face-to-face; (d) *Loss of Team Spirit* and trust among team members, and (e) *Cultural Differences* for when teams may involve members from different cultures.

Lanubile et. al. [23] identified seven standard collaboration tools used in GSE. Those primarily relevant to the way that team members operate and communicate and the monitoring of teams include 'Trackers', 'Build Tools', 'Modelers', 'Communication Tools', 'Web 2.0 Applications' and 'Project Management Tools'. Trackers manage issues as 'tickets', such as defects, changes and requests for support. Build tools allow for the creation and scheduling of workflows using safe and secure repository and build management and modelers include Management Systems that let members share explicit knowledge on the web. Communication tools include mainstream synchronous and asynchronous tools for members to communicate about the project and Web 2.0 applications extend on these previous communication tools and provide informal communication spaces for team members. Lastly, project management tools offer web-based interfaces to manage project information and milestones and can provide managers with an overview of the project status and information about team members and their activities. The authors [23] predicted that future SE tools will all include collaboration features. Similarly, in a systematic review of GSE tools, Portillo-Rodriguez et. al [22] identified a shift toward tools that are used to support communication aspects, such as Virtual Meeting Tools (12.2%), Software Engineering Management Tools (16%) and Knowledge Management Tools (16%). They suggest that the increase in these types of tools had been due to an increase of online communication methods used in software development, such as virtual communication, as well as the expansion of web-based tools for project control and the increasing use of wikis for knowledge sharing.

Although the number of tools has grown to accommodate online SE practices, the development or use of these tools for GSE have focused largely on providing affordances that alleviate issues in terms of communication and facilitation of team projects, e.g. by creating a sense of closeness among software engineers with the use of social network tools or by allowing members to collaboratively share knowledge with wiki tools. Limited attention is paid toward the development of tools or features for SE that monitor and visualise the team condition during tool usage in order to proactively alleviate common GSE issues [22].

In the field of education, there has been a growing interest in creating tools to help bring knowledge into practice in monitoring and assessing teamwork. A selected few are reviewed in this section according to their relevance to our study and the techniques they adopt. One of the earlier tools developed is ‘SPARK’, which is a move toward improving the confidentiality and ease of use in peer assessment through online submissions [24]. SPARK was created to improve the accuracy and fairness of assessment by providing confidential peer assessment opportunities. Other than confidentiality, another capability of SPARK is that the students have the ability to easily change their input during the semester. The tool was an improvement over the previous paper based methods of teamwork evaluation and can be considered as one of the early steps toward using technology to improve the assessment of teamwork.

‘SNAPP’ [25] is a tool that provides real-time analysis and visualisation of social networks between collaborators through the application of Social Network Analysis (SNA). SNAPP uses SNA to assess the cohesion of a team [26]. SNAPP can be used by educators or students as a way to reflect on team behaviour after a collaborative task has completed. Alternatively, educators can use the tool during collaborative discussions as a means to assist the early identification of isolated members or teams that do not appear to be forming cohesively. This tool can be useful for providing educators with information that may guide pre-emptive measures, however, educators have to collect data and process the tool, making it difficult to easily monitor multiple teams or acquire current information frequently.

‘Cohere’ is another tool that adopts network analysis of message topics in discourse in addition to SNA. Argument mapping is applied to collect and analyse discourse using message meta data [27]. For example, a message post will also include meta data on the target of the message and the underlying nature. This approach provides the opportunity to create a semantic network from otherwise temporal sequence of text messages. The authors have exploited this semantic network to analyse discourse networks and extract useful information such as hot and emerging topics and to identify people who are central to a discussion.

Sentiment analysis is another technique that has been applied to learning diaries to gain insight into students’ emotional experiences during the learning process [30]. Eight fundamental emotions of: ‘anger, trust, surprise, sadness, joy,

fear, disgust and anticipation’ [28] were tracked in the learning diaries and visualised as a spider graph [30]. This analysis can be useful as a means for lecturers to review learning tasks and gather information that might indicate if a particular student is having a negative experience and, therefore, if an intervention might be required.

A number of tools have focused on collecting and visualising access and activity data for education settings and Learning Management Systems. Such tools collect data about user access to course materials, the time spent viewing material as well as fine grained logging of user activity during quizzes, frequency of contributions in online discussion forums and so on [31], [32], [33], [25]. These tools have been available as plugins for learning management systems such as Moodle (www.moodle.org) or as stand alone applications.

III. MOTIVATION

Research has identified that with the increasing complexity and human-intensive nature of GSE models, GSE will require an increasing focus on tools and training that supports communication, evaluation and project management. This calls for a need for software engineers and researchers to develop innovative teamwork analysis tools that can support educators, SE managers and team members to make informed decisions about how to improve teamwork performance in current projects as well as for future projects.

IV. RESEARCH METHOD

The development of this dashboard extends on our previous work of analysing and exploring team role adoption and participation in online teamwork [11]. Previous work involved the development of a coding framework to analyse the team roles and behaviours [11] that could be used to guide manual content analysis of online discussions [34]. To develop the framework we used team role types defined by Dickinson and McIntyre [35] in their teamwork model, because these characterise roles within self-organising teams. The framework development also incorporated work from educational psychology to identify self-regulated learning behaviours [36], [37], [38] and previous work that explores argumentative knowledge construction [16], collaborative knowledge construction [39] and learning processes in collaborative environments [40]. Content analysis was applied to segments of discussion text, by coding the students’ text to a particular behaviour within each team role. The coded text could be comprised of one single word or two, a segment of text or sentences, depending on the interpreted behaviour.

The data that formed the basis for developing the dashboard was obtained from online teamwork discussions in one final-year undergraduate CS course at The University of Adelaide. The course explored complex problems inherent with computing operations carried out across a network, over more than one computer. Students in this level of study are expected to be able to apply their knowledge and to synthesise solutions to complex open-ended computing problems. The assessment used as the basis for our test case includes two

online collaborative tasks that contributed toward a modest 2% of the students' final grade. The instructor randomly allocated the 26 students to one of eight groups for the collaborative tasks. Students were asked to collaboratively respond to an open-ended problem for each of the two tasks by constructing a wiki as a team. Students were instructed to hold their team conversations using the Piazza (www.piazza.com) discussion feature, positioned underneath the wiki. The role of the collaborative tasks are to expose the students to difficult problems that require them to consider all of the possible issues that could occur during tasks carried out in this domain. The first task asked students how they would implement an Adaptive Timer System based on a previous assignment that used the Java Remote Method Invocation (RMI) System. Students were required to write a report to provide architecture for implementing the adaptive time on both the client and server side and consider the design issues. The second task invited students to devise a network design and discuss issues associated with the creation of a Massive Multi-User Online Roleplaying Game. The open-ended questions were selected as it is unlikely that one student holds all of the expertise in an area, which we hope encourages fluid transition of roles and the sharing of information between those who hold particular expertise at a point in time.

Manual content analysis [34] on the test case [11], revealed that students in the self-organised teams were self-initiating the adoption of some teamwork roles defined by Dickinson and McIntyre [35] but that the frequency of behaviours within the roles varied. Students were more commonly displaying behaviours associated with orientation, leadership and providing feedback. Similar to studies where leaders emerge in software organisation teams [17], one or two leaders usually emerged in the teams observed. A summary of behaviour frequencies and visualisations of team role behaviours using spider graphs revealed that while some members were more active, others were disengaged.

The content analysis of online discussion data required a great deal of human effort, in terms of coding, analysing, visualising and interpreting data. Such approaches would not be feasible for SE managers or educators wanting to efficiently monitor and evaluate online teams. Further, the time involved in content analysis only allowed for exploration of the team roles. There are a number of other interesting approaches for analysing online data as mentioned in the previous section, such as sentiment analysis and exploration of emotions, that, when combined in a dashboard, could provide greater insight into team condition and behaviour.

A. Research Questions

Our previous work leads us to investigate the following research question:

How might we develop a tool, based on previous work that automatically visualises role distribution and the emotional condition of a team?

In the following section, we present the dashboard. Firstly, we describe the technical information about the dashboard

development. Following, we explore how one would use and interpret the dashboard, testing it on the same data that was manually analysed as a test case.

V. THE TEAMWORK DASHBOARD

We present a prototype of an interactive dashboard solution that provides real-time feedback for SE team managers or educators. In Fig. 1 we present the dashboard in its entirety. The dashboard is comprised of a series of elements: team participation and role distribution, presented as spider graphs; team and individual sentiment analysis, presented as a line graph; and team and individual emotions, presented as spider graphs. Each element of the dashboard and its development will be elaborated on in more detail.

Discussion text is also displayed in a panel to the right, to provide the opportunity for the detailed inspection of particular parts of the discussion. Moving the 'time slider' in Fig. 5 or clicking a point in the sentiment chart in Fig. 4 shifts the discussion text window to focus on the discussion for the selected point in time and updates the visualisation. This allows the supervisor to inspect the discussion text that corresponds to critical points displayed in the graphs. As an example, a supervisor can select to inspect the discussion text for causation where a sentiment graph has started a downward trend. A visual representation of this procedure in use is displayed in Fig. 2. The indicator has been positioned at the start of the downward trend in the sentiment chart and the dashboard has shifted the text window to the corresponding message.

A. Team Role Behaviours

As identified in the literature, current tools provide little information about team role distribution and the self-organising roles adopted by team members over the duration of a project. To illuminate team role information, we incorporate team role distribution as a key element of our dashboard.

We chose to work with the teamwork roles defined by Dickinson and McIntyre [35] because their teamwork model focuses on the practical roles of self-organising teams. The authors classified seven core components of teamwork: 'Team Leadership', 'Team Orientation', 'Monitoring', 'Coordination', 'Communication', 'Feedback' and 'Backup Behaviour'. Additionally, we divided Dickinson and McIntyre's roles of 'Feedback' and 'Backup Behaviour' into their sub-categories for analysis as 'seeking' and 'providing' behaviours are different in their purpose and we wanted to know when students were providing support or seeking support.

The process of automating the identification of behaviours for the dashboard, based on previous manual content analysis of team roles [11] involved identifying common keywords and phrases in the coded text for each of the teamwork roles. The text was manually explored in NVivo 10 using keyword frequency searches and by reviewing the text that had been coded to identify common phrases and keywords for each role. These were noted in a document and the collection of search terms and phrases were tested and refined in NVivo 10 using

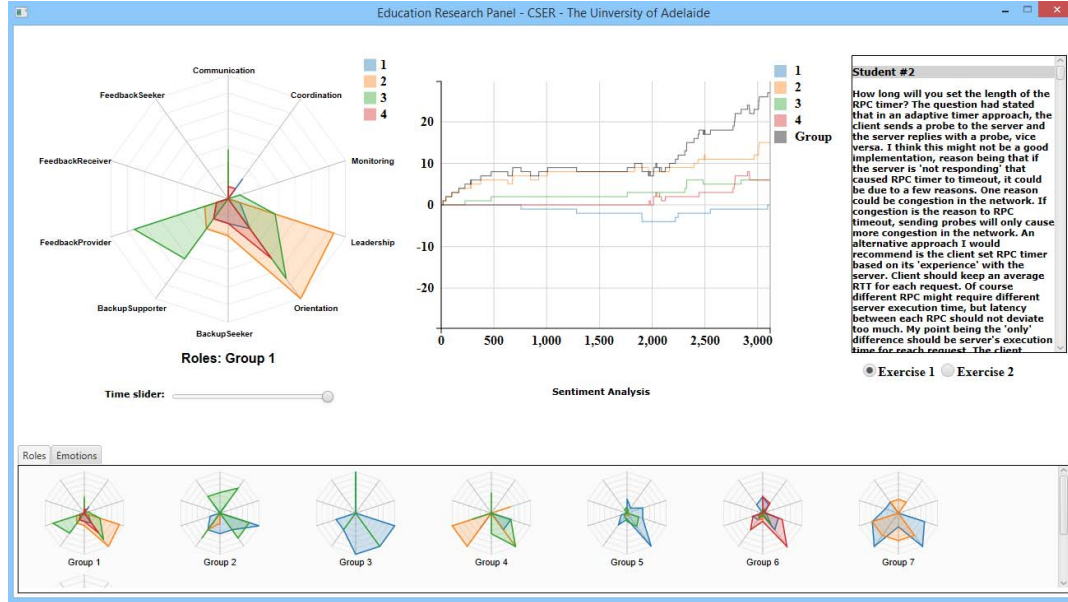


Fig. 1: Teamwork analysis dashboard

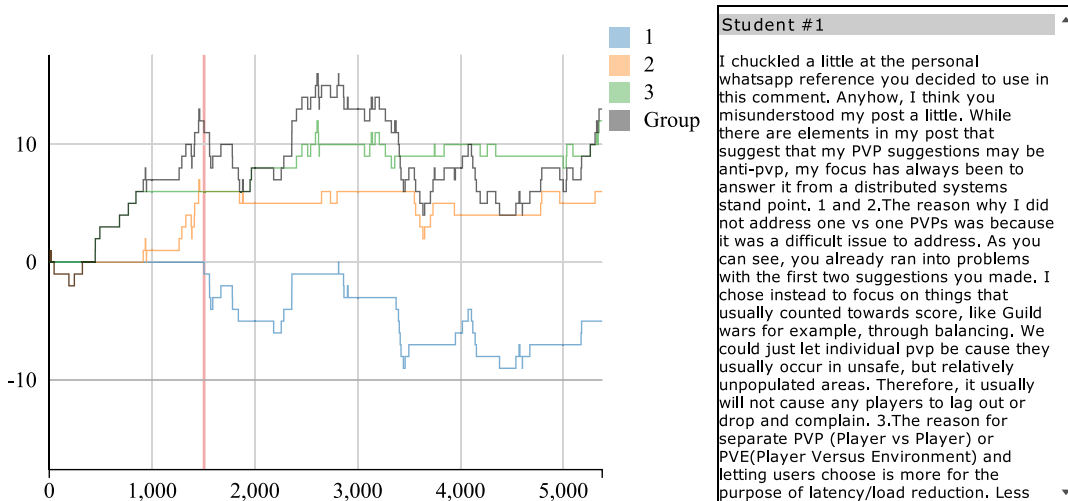


Fig. 2: Investigation of a downward trend

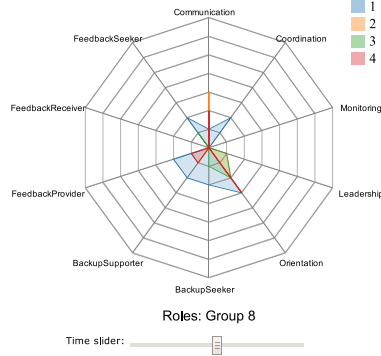
search queries. Words or phrases that retrieved text that were deemed outside of the scope of the role were removed. Once the researcher was satisfied with the closeness of the search retrieval, the document was used by the software engineer as a guide for the automation in the learning analytics dashboard.

The list of patterns used for each role is then converted into Perl/Java style regular expressions. The following is an example of how the regex list would look like:

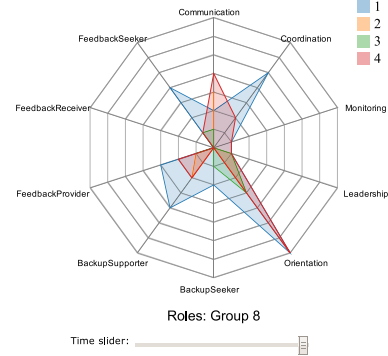
```
RegexBank=
["Leadership":
  "\b(we|you) \b\s*(\b(should|must|ought to
    |better|need to)\b) |\b(I) ('m|\s*\bam)?\b
```

```
\s*\b(expect (ing)? \b|...",
"Communication":
  "\b(I) ('d|\s*\bwould)?\b\s*\b(like(\s*
    to)? \b|\b(I)\b\s*(\b(guess|think|don't|
    do not)?\s*(understand|know|...")
  ...
```

Role vectors are calculated for each message in the discussion with each component representing one of the roles. The presence of matches for each pattern in a messages would add to the value of the corresponding role in the role vector for that message. For example, a brief pseudocode of this operation is as follows:



(a) Halfway through the project



(b) End of the project

Fig. 3: Visualization of teamwork roles

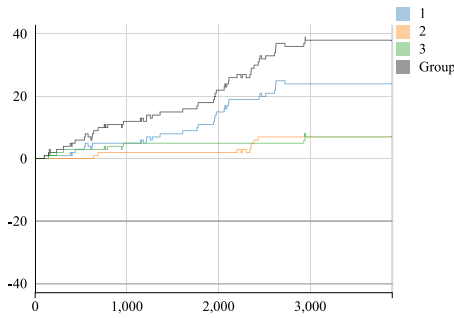


Fig. 4: Positive/negative sentiment

```

Roles=[
"Communication", "Coordination",
"Monitoring", "Leadership",
"Orientation", "Backup Seeker",
"Backup Supporter", "Feedback Provider",
"Feedback Receiver", "Feedback Seeker"]

RoleVector=0
FOR message IN messages
  FOR role IN Roles
    IF matches(message.content
      ,RegexBank[role])
      RoleVector[role]+= 1
    ENDIF
  RolesHistory[message.time]=RoleVector

```

Figure 3 presents a spider graph for roles adopted by each member in a team of four at two points of time: the middle of the discussion (Fig. 3a) and at the end of the discussion (Fig. 3b).

B. Sentiment Analysis

Figure 4 presents a panel that displays the positive and negative sentiment of the team contributions. The line chart represents the sentiment of individual team members as well as the total sentiment of the team from the beginning of the

project. The frequency of positive and negative terms is used along with a list of ‘stop words’ to determine sentiment [41]. The sentiment bar starts at a neutral position at the beginning of the project. The line graph then increases for each positive emotion expressed in the text and decreases for each negative term encountered. This captures and visualises both negative and positive aspects of a piece of discussion text. Rather than providing a per-message sentiment analysis which analyses a whole message for a number indicating the mood of the message [41], we used the same word lists but adopted this method so that we can visualise trends of positivity/negativity as early as possible. The interactive line graph allows the supervisor to click on trending points of interest and to see the accompanying discussion text.

Skipping certain domain specific words might be necessary depending on the discussion domain. For example, the word ‘kill’ in computing does not bear the negative sentiment that it would otherwise have in some other domains. A ‘skip list’ was prepared, including commonly used domain specific words which would have resulted in false negatives. The dashboard skips any negative sentiment resulting from one of the terms included in this file to ensure minimal false negative results.

C. Team Emotion

To compensate for the loss of interpersonal cues in online or distant teams, we have developed a feature that provides analysis and visualisation of emotions in realtime for team members and teams as a whole. Eight basic emotions of ‘anger, anticipation, disgust, fear, joy, sadness, surprise, trust’ [28] are mined from member contributions to the discussions. To extract individual emotions the contributions are matched against the ‘NRC Word Emotion Lexicon’ [29]. This word emotion lexicon has been used before in extracting student emotions from their learning diaries [30]. We improve the versatility by pairing term frequency for each word with emotional value of the word to create an emotion vector for each piece of discussion text contributed by a team member. The overall emotion vector in any given point of time will be the accumulation of emotion vectors extracted for messages

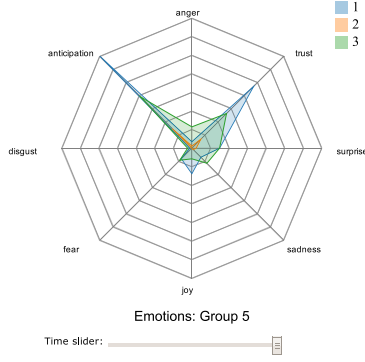


Fig. 5: Visualisation of accumulated emotions E_t

from the beginning of the discussion until the given time. To avoid ambiguity a mathematical representation of this value is shown in Equation 1. This enables the supervisor to not only view the current emotional condition of the team, but to also move the interactive time slider to any past point in time to observe the evolution of emotions. Per message calculation of emotion vectors also provides the opportunity for parallel computation as each message is analysed separately.

$$E_t = \sum_{i=0}^t V_i \quad (1)$$

E_t : Accumulated Emotions at time t

V_i : Emotion Vector for message posted at time i

The emotions are visualised as a spider graph in which the axes correspond to emotions and each colour represent a team member. A sample screenshot of the emotion graph for a team is provided in Fig. 5. The graph represents the point in time for which the emotional condition of the team is being visualised, according to the position of the 'time slider'. Shifting the time slider will update the graph. Moving the time slider from one end to the other presents an animated visualisation of emotions as they evolve during the collaborative task.

VI. CASE STUDY PRESENTATION OF THE DASHBOARD: INTERPRETING TEAMWORK DATA

We imported the Piazza data from the 'Distributed System' course discussions into the database. The data comprised of collaborative problem solving discussions for two carefully designed open-ended problem-solving tasks. In the first task the question is presented as a dry, technical problem for students to solve. The second task is framed around an issue affecting Massive Multiplayer Online Game systems.

Figures 6 and 7 display a view of team roles and tasks adopted by students during the course. The thumbnails on the left visualise team roles adopted, with axes corresponding to the roles of 'Communication', 'Coordination', 'Monitoring', 'Leadership', 'Orientation', 'Backup Seeker', 'Backup Supporter', 'Feedback Provider', 'Feedback Receiver' and 'Feedback Seeker' in a clockwise manner. The thumbnails on the right visualise the emotional climate of the teams, with

axes corresponding to each emotion of 'anger, trust, surprise, sadness, joy, fear, disgust, anticipation' in a clockwise manner from the top.

To use the tool, educators or team managers would examine the various spider graphs across all teams. One would look for irregular or unusual patterns in either the team role or emotion graphs. In a comparison of the results of analysis of team roles by the tool, with manual analysis of team roles [11] the only difference we saw was that a few more team roles were detected by the tool which had escaped manual analysis and vice versa. However the role identification pattern depicted in the spider graphs were similar for both manual and automated tool analysis.

The students remained in the same teams for both tasks and both tasks contributed toward the same (2%) mark. By comparing the emotion charts between the two tasks in Fig. 6b and Fig. 7b, an observation is that more emotions were expressed by the students in the second task. Another interesting observation by comparing the role charts in Fig. 3a and Fig. 3b across both tasks shows that some students tended to show more involvement in team roles in the second task. For instance in 'Group 1', one of the students (colour-coded green) shows a change in their team role pattern and overtakes the leadership and another student (colour-coded red) covers more team roles. In our deductions we hypothesise that this may be due to the nature of the open-ended questions. The first question was quite dry in nature, whereas the second question involved students drawing on their experience and knowledge (or lack of knowledge) of massive multiplayer games. Future research could use a tool such as this to compare between types of roles and emotions that emerge between different types of task questions. Educators might find this particularly useful when reviewing online learning activities, with the goal to promote active discussion and engagement, and to drive intrinsic interest.

Another feature of the tool is the ability for the user to interact with the 'time sliders' and sentiment analysis graph. Progression of team roles and emotions among members or the team as a whole can be easily viewed by using the time slider. In our manual analysis, we perceived that teams began by developing through a phase of orientation toward the task, in which leadership roles emerged [11]. Afterwards, the teams would progress through the discussion, displaying various roles as required. In exploring if this was the case with the dashboard tool, we used the time slider. By setting the time slider to the first 25% span, we also saw that most teams typically had an initial phase where team roles mostly featured orientation and a leader or two.

Many of our initial observations and results, identified through manual content analysis, were confirmed with the exploration and use of the teamwork dashboard tool. The automated analysis produced similar results to the manual analysis, however the tool provided a much more efficient approach that allows the user to spend less time collecting and analysing the teamwork and more time exploring teamwork activity and team progression through the tasks.

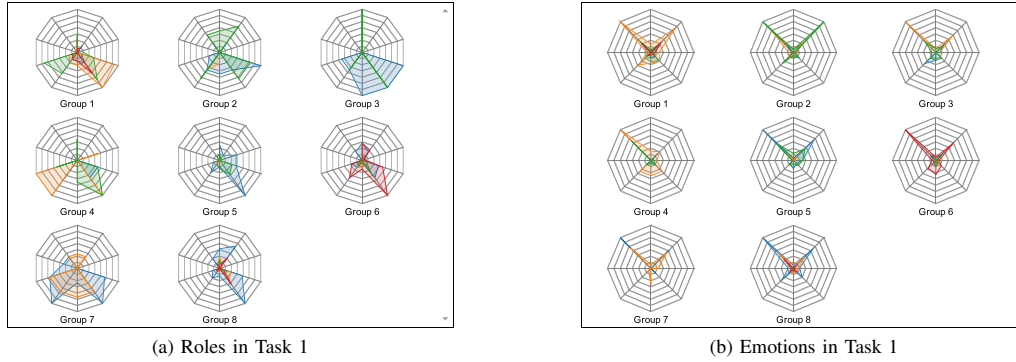


Fig. 6: Roles and emotions in Task 1

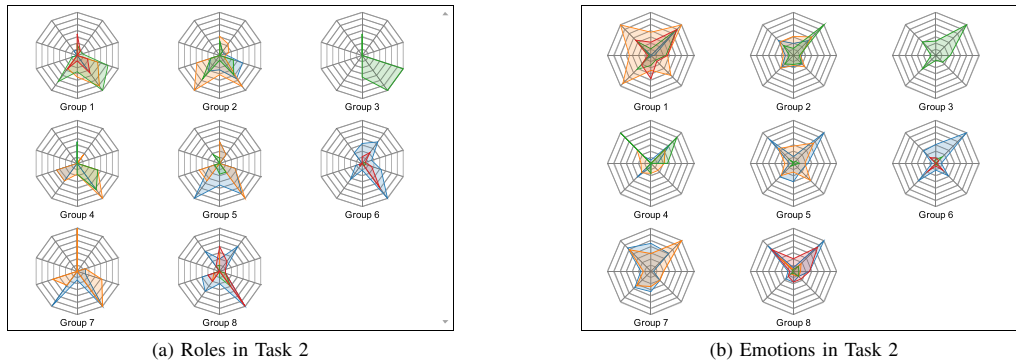


Fig. 7: Roles and emotions in Task 2

VII. LIMITATIONS AND FUTURE RESEARCH

A limitation of the tool is that it is based on term frequency and phrases detected in discussion text pertaining to certain roles, emotions and sentiment. Particular team members may appear to be inactive according to the frequency of behaviours presented in the dashboard but actually be contributing technical information. We are able to measure contributions by word count, but are not at the stage of measuring the quality of contribution. However, the focus of the dashboard was to move beyond measures of contributions and extend on previous work by providing a fresh angle on how teams are functioning, pertaining to aspects relevant to teamwork as these are desired skills by SE employers [1], [4]. This tool would also provide a unique perspective that complements existing tools, where educators and SE managers may be able to reflect on both contributions and activity logs, in conjunction with elements presented in the teamwork dashboard.

This tool has been developed and tested on the dataset from two collaborative tasks from one university SE course. Although this is a small case study, it allowed for the researcher and software engineer to compare and contrast the manual analysis and automated approaches. One particular aspect unique to the SE context is the use of words that are inherently tied to negative sentiment. This is often due to

the way in which software engineers discuss complex issues in software development to identify solutions and improve products. Currently, the sentiment analysis in the dashboard tool skips any negative sentiment resulting from one of the terms included in the 'stop list' file to ensure minimal false negative results. Future work will augment this approach by incorporating word-sense disambiguation. Future work will also involve testing the dashboard on other disciplines, courses and environments (besides Piazza) so that we are able to refine and evaluate the processes for automatically analysing and visualising teamwork behaviours and conditions. Moreover, this work offers opportunities to explore different ways of presenting the information to address educator and team manager needs and for exploring new techniques that can be integrated into the dashboard.

VIII. CONCLUSION

This work contributes to the field of learning analytics and the development of tools for software engineering and teamwork. This teamwork dashboard brings together existing techniques with new approaches that are grounded in educational theory. This tool may assist educators and team managers with more efficient and effective measures of monitoring online teamwork and also complements existing communication tools

adopted by both industry and education. The development of this tool provides a new avenue for research and opens opportunities for expanding and assessing the tool's features to align with educator and team manager needs.

REFERENCES

- [1] W. Archer and J. Davison, "Graduate employability: what do employers think and want?," *Graduate Employability: The Views of Employers*, The Council for Industry and Higher Education, London, UK, 2008.
- [2] J. Verner, O. Brereton, B. Kitchenham, M. Turner and M. Niazi, "Risks and risk mitigation in global software development: A tertiary study," *Information and Software Technology*, vol. 56, no. 1, pp.5478, 2014.
- [3] G. Ruhe and C. Wohlin, "Software Project Management: Setting the Context," *Software Project Management in a Changing World*, Springer Berlin Heidelberg, pp.1-24, 2014.
- [4] M. Robles, 2012, "Executive perceptions of the top 10 soft skills needed in today's workplace," *Business Communication Quarterly*, October 8, 2012.
- [5] A. Begel, A and B. Simon, "Novice software developers, all over again," paper presented at Workshop on Computing Education Research, Sydney, Australia, 2008a.
- [6] A. Begel, A and B. Simon, "Struggles of new college graduates in their first software development job," paper presented at SIGCSE Technical Symposium on Computer Science Education, Portland, USA, 2008b.
- [7] A. Radermacher and G. Walia, "Gaps between industry expectations and the abilities of graduates," *Technical Symposium on Computer Science Education*, Denver, Colorado, pp.525-530, 2013.
- [8] S. Ruff and M. Carter, "Communication learning outcomes from software engineering professionals: a basis for teaching communication in the engineering curriculum," *Frontiers in Education Conference*, San Antonio, Texas, pp.1-6, 2009.
- [9] K. Falkner and N. Falkner, "Supporting and structuring "Contributing Student Pedagogy" in computer science curricula," *Computer Science Education*, vol. 22, no. 4, pp.413- 443, 2012.
- [10] R. Lingard and S. Barkataki, S, "Teaching teamwork in engineering and computer science," *Frontiers in Education Conference*, Rapid City, South Dakota, pp.F1C-1-F1C-5, 2011.
- [11] R. Vivian, K. Falkner and N. Falkner, "Analysing computer science students' teamwork role adoption in an online self-organised teamwork activity," pp.105-114, *Koli Calling*, Finland, November, 2013.
- [12] A. Powell, G. Piccoli and B. Ives, "Virtual teams: a review of current literature and directions for future research," *SIGMIS Database*, vol. 35, no. 1, pp.6-36, 2004.
- [13] R. Hughes and S. Jones, "Developing and assessing college student teamwork skills," *New Directions for Institutional Research*, vol. 2011, no. 149, pp. 53-64, 2011.
- [14] L. J. ChanLin and K. C. Chan, "Group learning strategies for online course," *Procedia - Social and Behavioral Sciences*, vol. 2, no. 2, pp.397-401, 2010.
- [15] A. El-Abbassy, R. Muawad and A. Gaber, "Evaluating agile principles in CS Education," *International Journal of Computer Science and Network Security*, vol. 10, no. 10, pp.19-28, 2010.
- [16] G. Weinberg, *The psychology of computer programming*. Computer Science Series, New York, United States: Van Nostrand Reinhold Company, 1971.
- [17] J. W. Strijbos, M. De Laat, R. Martens and W. Jochems, "Functional versus spontaneous roles during CSCL," paper presented at Computer Support for Collaborative Learning, Taipei, Taiwan, 2005.
- [18] J. Hamer, Q. Cutts, J. Jackova, A. Luxton-Reilly, R. McCartney, H. Purchase, C. Riedesel, M. Saeli, K. Sanders and J. Sheard, "Contributing student pedagogy," *Special Interest Group on Computer Science Education (SIGCSE) Bull.*, vol. 40, no. 4, pp.194-212, 2008.
- [19] C. Loftus, L. Thomas, C. and Zander, "Can graduating students design: revisited," *Technical Symposium on Computer Science Education*, Dallas, Texas, pp. 105- 110, 2011.
- [20] K. Falkner, N. J. G. Falkner, and R. Vivian, "Collaborative Learning and Anxiety: A Phenomenographic Study of Collaborative Learning Activities," in *Proceeding of the 44th ACM Technical Symposium on Computer Science Education*, New York, USA, 2013, pp. 227-232.
- [21] B. Oakley, D. Hanna, Z. Kuzmyn and R. Felder, "Best practices involving teamwork in the classroom: results from a survey of 6435 Engineering student respondents," *IEEE Transactions on Education*, vol. 50, no. 3, pp. 266- 272, 2007.
- [22] J. Portillo-Rodríguez, A. Vizcaino, M. Piattini and S. Beecham, "Tools used in Global Software Engineering: A systematic mapping review," *Information and Software Technology*, 54, 663-685, 2012.
- [23] F. Lanubile, C. Ebert, R. Prikladnicki and A. Vizcaino, "Collaboration tools for Global Software Engineering," *IEEE Software*, vol. 27, no. 2, pp.5255, 2010.
- [24] M. Freeman and J. McKenzie, "SPARK, a confidential web-based template for self and peer assessment of student teamwork: benefits of evaluating across different subjects," *British Journal of Educational Technology*, vol. 33, no. 5, pp. 551-569, Nov. 2002.
- [25] A. Bakharia and S. Dawson, "SNAPP: A Bird'S-eye View of Temporal Participant Interaction," in *Proceedings of the 1st International Conference on Learning Analytics and Knowledge*, New York, USA, 2011, pp. 168-173.
- [26] C. Reffay and T. Chanier, "Social Network Analysis Used for Modelling Collaboration in Distance Learning Groups," in *Proceedings of the 6th International Conference on Intelligent Tutoring Systems*, London, UK, 2002, pp. 31-40.
- [27] A. De Liddo, S. B. Shum, I. Quinto, M. Bachler, and L. Cannavacciuolo, "Discourse-centric Learning Analytics," in *Proceedings of the 1st International Conference on Learning Analytics and Knowledge*, New York, USA, 2011, pp. 23-33.
- [28] Plutchik, Robert. "A general psychoevolutionary theory of emotion," *Approaches to emotion*, 1984, ch. 8, pp. 197-219.
- [29] S. M. Mohammad and P. D. Turney, "Crowdsourcing a Word-Emotion Association Lexicon," *Computational Intelligence*, vol. 29, no. 3, pp. 436-465, 2013.
- [30] M. Munezero, C. S. Montero, M. Mozgovoy, and E. Sutinen, "Exploiting Sentiment Analysis to Track Emotions in Students Learning Diaries," in *Proceedings of the 13th Koli Calling International Conference on Computing Education Research*, New York, USA, 2013, pp. 145-152.
- [31] J. Jovanović, D. Gašević, C. Brooks, V. Devedžić, and M. Hatala, "LOCO-Analyst: A Tool for Raising Teachers' Awareness in Online Learning Environments," in *Creating New Learning Experiences on a Global Scale*, E. Duval, R. Klamma, and M. Wolpers, Eds. Springer Berlin Heidelberg, 2007, pp. 112-126.
- [32] L. P. Macfadyen, and P. Sorenson. "Using LiMS (the Learner Interaction Monitoring System) to track online learner engagement and evaluate course design." In *Proceedings of the 3rd international conference on educational data mining*, Pittsburgh, USA, pp.301-302, 2010.
- [33] D. Petkovic, et al. "Work in progress-e-TAT: Online tool for teamwork and "soft skills" assessment in software engineering education." *Frontiers in Education Conference (FIE)*, IEEE, Arlington, VA, 2010.
- [34] Y. Zhang and B. Wildermuth, "Qualitative analysis of content," in *Applications of Social Research Methods to Questions in Information and Library Science*, B. Wildermuth (Ed), Libraries Unlimited: Westport, pp.308-319, 2009.
- [35] Dickinson, Terry L., and Robert M. McIntyre. "A conceptual framework for teamwork measurement," *Team performance assessment and measurement*, 1997, pp. 19-43.
- [36] B. Zimmerman, "Becoming a self-regulated learner: which are the key subprocesses?," *Contemporary Educational Psychology*, vol. 11, no. 4, pp. 307- 313, 1986.
- [37] B. Zimmerman and M. Pons, "Development of a structured interview for assessing student use of self-regulated learning strategies," *American Educational Research Journal*, vol. 23, no. 4, pp.614-628, 1986.
- [38] A. Hadwin, S. Järvelä and M. Miller, "Self-regulated, co-regulated, and socially shared regulation," in B. Zimmerman and D. Schunk (eds), *Handbook of self-regulation of learning and performance*, Taylor & Francis, New York, pp.65-63, 2011.
- [39] C. Hmelo-Silver, "Analyzing collaborative knowledge construction: multiple methods for integrated understanding," *Computers & Education*, vol. 41, no. 4, pp.397-420, 2003.
- [40] C. Hmelo-Silver, E. Chernobitsky, and R. Jordan, "Understanding collaborative learning processes in new learning environments," *Instructional Science*, vol. 36, no. 5, pp.409-430, 2008.
- [41] M. Hu and B. Liu, "Mining and Summarizing Customer Reviews," in *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, New York, USA, 2004, pp. 168-177.